

Machine Learning (EE526/RA506)

End-Semester Exam

Date: 30/04/2024

Time: Three Hours

Full Marks: 160

[Q1] Consider the case of incremental mean vector and covariance matrix update for a streaming data problem. The respective mean vector and covariance matrix from the previous instant $(t - 1)$ are μ_{t-1} and C_{t-1} . Deduce the updated values of μ_t and C_t with present data vector x_t and a fixed update rate α . [2+6=8]

[Q2] The dataset $S = \{x_i; x \in \mathbb{R}^d; i = 1, \dots, n\}$ ($n = 210000, d = 10$) is subjected to hierarchical two-means clustering. This is realized through a fully-grown binary tree which terminates with at least $m = 100$ points at each leaf node. The root node is considered to be at depth $D = 0$. (a) What value of maximum depth D_{max} should be set for constructing this tree? (b) Estimate the total number of bytes required for parametrizing all the split nodes of the fully-grown tree of depth D_{max} . Note that both integers and floating-point numbers use 4 bytes. (c) How many clusters will be formed with this value of D_{max} and a fully-grown tree? (d) We want to classify an unseen instance $u \in \mathbb{R}^d$ with one of these cluster labels. How many floating-point computations will be required for this classification? [2+4+2+4=12]

[Q3] Consider the following dataset D which is subjected to mean-shift clustering through mean-shift iterations using an Epanechnikov kernel of band-width $h = 1.73$. Let the cluster mode value at the k^{th} iteration be $x^{(k)} = \begin{bmatrix} 3.7 \\ 2.5 \end{bmatrix}$. Evaluate the mode value $x^{(k+1)}$ at the $(k+1)^{th}$ iteration. [10]

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
4.2	2.5	4.7	3.4	2.9	4.3	6.0	2.6	3.1	1.9
1.7	0.4	2.0	4.6	1.5	0.9	2.8	2.2	4.8	3.9

[Q4] The parameters for four classes are as presented in the following table. The likelihood of each class is modelled using a single Gaussian (in 2D feature space) parameterized by its respective mean (μ_1, μ_2), standard deviations (σ_1, σ_2) and prior π . Use the MAP rule to classify the unseen instance $x = (-2.5, 2.5)$. [20]

	Class-1	Class-2	Class-3	Class-4
μ_1	-5	0	-2	-1
μ_2	5	0	2	1
σ_1	1	5	3	1
σ_2	1	5	3	1
π	0.3	0.25	0.25	0.2

[Q5] Consider a multi-category classification problem that led to the following confusion matrix. Report the Class-wise F1 scores. [3 x 5 = 15]

Multi-Category Confusion Matrix		Classification Results				
		C1	C2	C3	C4	C5
Ground Truth	C1	23	51	11	2	31
	C2	41	23	13	35	5
	C3	3	13	81	6	7
	C4	12	52	11	25	9
	C5	61	27	1	3	11

140 166 117 21 63

[Q6] The 2D input $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is related to the output 1D input y in the following manner.

$$u = \omega_1 x_1 + \omega_2 x_2$$

$$y = (1 + e^{-u})^{-1}$$

A few values of x and corresponding values of y are tabulated as follows.

x_1	-3.5	2.1	-4.3	1.7	0.1
x_2	1.9	-3.3	5.1	2.2	-0.1
y	0.21	0.01	0.97	0.98	0.45

Estimate the values of ω_1 and ω_2 .

[20]

[Q7] Suggest a ML model that can perform both Logistic Regression and Linear Regression with certain modification. Explain your answer. [5]

[Q8] Consider a binary classification problem with training dataset $D = \{(x_i, y_i); i = 1 \dots n\}$ where $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$. The classifier operates on the input in the following manner: $u_i = x_i^T A x_i + b^T x_i + c$ and $\hat{y}_i = \text{Sigmoid}(u_i)$ where, $A \in \mathbb{R}^{d \times d}$, $b \in \mathbb{R}^d$ and $c \in \mathbb{R}^1$ are the parameters of the classifier. The total error is defined as $E = \frac{1}{n} \sum_{i=1}^n e_i^2$ where, $e_i = y_i - \hat{y}_i$. This total error E is minimized through gradient descent. Accordingly, deduce the parameter update ($A_{ij}^{(t)}$, $b_i^{(t)}$, $c^{(t)}$) equations. [15]

[Q9] A Single Layer Feedforward Network (SLFN) has $D = 2$ inputs, $Q = 3$ nodes in hidden layer and $M = 2$ outputs. It uses sigmoid activation functions in both hidden and output layers. The weights and biases for the SLFN are as follows.

$$W_1 = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix} \quad B_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix} \quad B_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix}$$

Evaluate SLFN output Y for the input $X = [-0.25 \quad 0.57]^T$ [20]

[Q10] Consider the MLP with input layer X (D nodes), single hidden layer H (Q nodes) and output layer Y (M nodes). The respective weight matrices and bias vectors are (W_1, B_1) for X to H and (W_2, B_2) for H to Y . Deduce expressions for weight and bias update for minimizing the cross-entropy loss over the training dataset $\{(X_n, Y_n); n = 1, \dots, N\}$. The output layer nodes host the *SoftMax* activation function. The hidden layer nodes host the activation function $g(u) = \tanh(u)$. [20]

[Q11] Consider a two-category data set D , containing $q\%$ positives. State the expressions for *Misclassification*, *Entropy* and *Gini* impurity measures for this problem. What value of q will maximize these impurity measures? Plot the graphs of these three impurity measures with respect to q . [15]

----- All the Best -----